

Life Table & Mortality Analysis

Vietnam Study 2000 and 2021

WHO Global Health Observatory

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Summary

This report documents a complete actuarial mortality analysis for the Vietnamese population, using WHO Global Health Observatory data from 2000 and 2021. To do this, we developed a Python-based life table model to derive key actuarial functions (q_x , l_x , d_x , ${}_nL_x$, T_x , e_x) under the UDD assumption. The analysis highlights three findings with direct pricing relevance: a sharp spike in young male mortality around age 20 (3.37× risk ratio), driven largely by accidents; an increasing concentration of deaths at older ages as life expectancy rises; and a steady 20-year overall mortality improvement. Together, these structural shifts create divergent risk implications for term life versus annuity products.

1. Objective & Scope

This project builds an abridged life table for Vietnam using raw WHO mortality data, evaluating the results through the lens of life insurance pricing. By comparing data from 2000 and 2021, the analysis tracks how Vietnam's mortality profile has evolved over two decades of rapid economic development, highlighting what these shifts mean for product design.

Please note that the scope is strictly limited to population-level mortality. Since no policyholder experience data was available, these tables represent a general population baseline rather than an insured lives study. Consequently, all pricing implications discussed in Section 4 are directional and should not be taken as calibrated metrics for a specific product.

Life table functions computed:

- q_x and p_x — probability of death and survival within each age interval
- l_x and d_x — survivors and deaths from a radix cohort of 100,000
- ${}_nL_x$ — person-years lived in $[x, x+n)$, under the UDD assumption

- T_x and e_x — total future person-years and curtate life expectancy at age x

2. Data & Methodology

2.1 Data

Mortality data were retrieved programmatically from the WHO GHO REST API (indicator LIFE_0000000030, country code VNM). The indicator provides abridged q_x values by sex and five-year age group for multiple calendar years. Year 2021 was selected as the most recent available; year 2000 was fetched separately for the temporal comparison in Section 3.3. All 19 age groups (0, 1–4, 5–9, ..., 85+) were present for both years and both sexes, with no missing values after extraction.

One data quality check was applied before modelling: all q_x values were verified to lie strictly within $[0, 1]$. The terminal group (85+) was set to $q_{85} = 1.0$ by construction, ensuring the cohort is fully extinguished at the open end.

2.2 Model

The life table engine was implemented as a Python class (AbridgedLifeTable), taking an age array and q_x vector as inputs. The radix was fixed at $l_0 = 100,000$. Person-years were computed under the **Uniform Distribution of Deaths (UDD)** assumption:

Function	Formula	Assumption
${}_nL_x$ (interior groups)	$n \cdot (l_x - 0.5 \cdot d_x)$	UDD within $[x, x+n)$
${}_{\infty}L_{85}$ (open tail)	$l_{85} \times e_{85}$	$e_{85} = 5.5$ yrs (WHO-consistent)
T_x	$\Sigma {}_nL_x$ (reverse cumulative sum)	—
e_x	T_x / l_x —	—

Sensitivity of e_0 to the open-tail assumption is reported in Section 4.

3. Results

3.1 Summary — Vietnam 2021

Metric	Male	Female	Gap (F – M)
e_0 (life expectancy at birth)	69.95 yrs	78.19 yrs	+8.24 yrs
e_{40}	33.34 yrs	40.28 yrs	+6.94 yrs
e_{65}	13.94 yrs	18.09 yrs	+4.15 yrs

l_{65} (survivors to age 65)	~71,500	~84,200	+12,700
Peak Risk Ratio (q_M/q_F)	Age 20 — 3.37×	—	—

3.2 Mortality Curve, Survival Curve, and Risk Ratio (2021)

Figure 1 illustrates the mortality and survival profiles for Vietnam in 2021, highlighting two critical structural trends:

The accident hump.

While the female mortality q_x curve follows a steady, predictable Gompertz trajectory from age 10 onwards, the male curve sharply diverges between ages 15 and 25. The data shows that at age 20, male mortality risk is 3.37 times higher than that of females, before gradually stabilizing to a 2.5× multiple by age 35. The sheer magnitude of this elevation dictates that any term life pricing model targeting young men must explicitly account for this excess risk.

Curve Rectangularization (Concentrated Old-Age Mortality)

The survival l_x data reveals a clear "rectangularization" effect: over 90,000 individuals (per 100,000) survive past age 50, followed by a steep drop-off in older age groups. The practical consequence is that mortality risk in Vietnam is now heavily concentrated in the 60+ bracket. For actuaries, this shift has profound implications for product design and reserve adequacy, particularly in long-duration contracts and annuities.

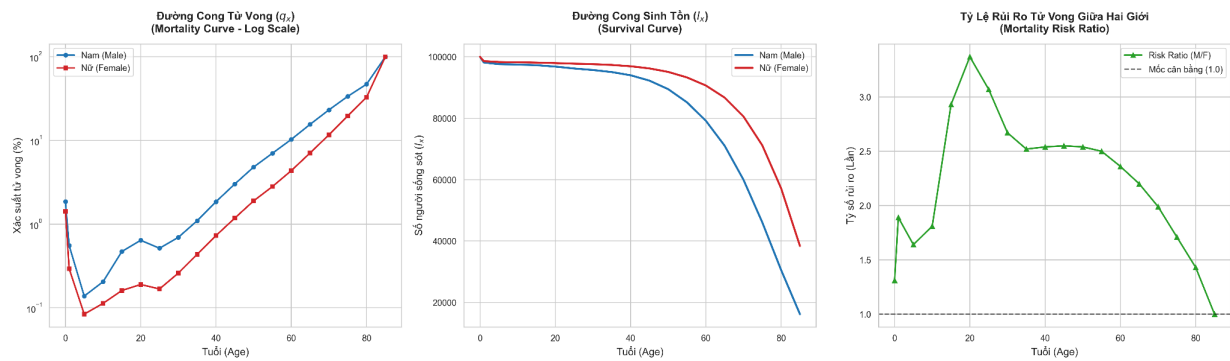


Figure 1. Left: q_x by sex on a log scale (Vietnam 2021). Centre: survival curve l_x . Right: gender risk ratio q_M/q_F by age.

3.3 Temporal Comparison — 2000 vs 2021

Comparing the mortality profiles across a 20-year horizon reveals a substantial, systematic improvement in Vietnamese longevity. The downward shift in mortality q_x is most pronounced in two specific demographics: infants/early childhood (ages 0–5) and working-age adults (ages 40–65). These gains directly reflect broader national advancements, including improved maternal-infant healthcare infrastructure and better management of cardiovascular risks among the working population. Parallel to the falling mortality rates, the survival curves exhibit a steady upward and rightward expansion. This extension is particularly visible from age 60 onwards, signaling a clear structural shift toward an aging population. For insurance portfolios, this historical trajectory emphasizes that longevity risk is no longer a theoretical concern but an active trend that must be priced into long-term annuity and pension products.

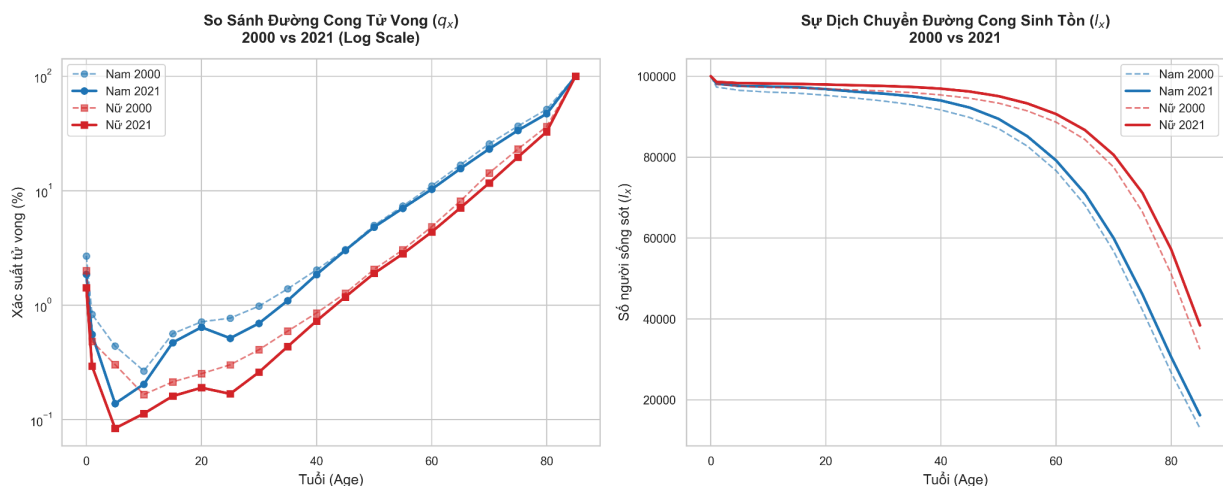


Figure 2. Mortality and survival curves for Vietnam, 2000 (dashed) vs 2021 (solid). Male: blue. Female: red.

4. Actuarial Implications

4.1 Gender Risk Differentiation in Term Life Pricing

With the male-to-female risk ratio continuously exceeding $2.5\times$ between ages 20 and 35, adopting a gender-neutral premium rate introduces a significant cross-subsidization risk. For instance, in a standard 10-year term life policy issued at age 25, a blended rate would materially overcharge female policyholders while undercharging males. Given this stark disparity, implementing gender-differentiated pricing becomes an actuarial necessity to ensure fairness and prevent adverse selection, rather than just a commercial preference.

4.2 Mortality Improvement: Divergent Impact by Product Line

The comparison between 2000 and 2021 data highlights a remarkable increase in life expectancy at birth e_0 , rising by approximately 8 years for males and 11 years for females. While this long-term trend is a positive socio-economic indicator, it creates contrasting financial dynamics across different insurance product lines:

For Term Life Portfolios (A Mortality Profit): Sustained reductions in q_x values mean fewer expected death claims in the near term. Insurers pricing a term product today using the 2021 benchmark will likely experience a comfortable mortality surplus if these trends hold true. However, the risk here is primarily commercial rather than financial: competitors utilizing the most up-to-date, improved mortality data can undercut premiums and capture market share.

For Annuity and Pension Products (A Longevity Risk): The dynamic completely reverses. Lower mortality directly translates to prolonged payout periods. For instance, at age 65, female life expectancy e_{65} already reaches 18.09 years under 2021 mortality—a substantial 4.15-year extension compared to 2000. Companies that continue to price lifetime annuities using outdated 2000-era tables face severe under-reserving risks, as cumulative payouts will outlast original pricing assumptions and strain solvency.

4.3 Static Tables and the Case for Improvement Factors

The 20-year gap between this project's two reference years illustrates why locking in a static mortality table is not appropriate for long-duration products. The standard actuarial response is to incorporate **mortality improvement scales** — factors that project q_x forward year by year — rather than assuming

today's rates persist unchanged. For products with 20–30 year payout horizons, the difference between a static and a projected table is not a second-order refinement; it is a first-order solvency consideration.

4.4 Sensitivity to the Open-Tail Assumption

To evaluate the robustness of our model, the terminal life expectancy parameter e_{85} was varied across a plausible range. Table 2 outlines the resulting impact on life expectancy at birth e_0 for males:

e_{85}	e_0 Male	e_0 Female	Change vs base
4.0 yrs (low)	69.82	78.05	−0.13 yrs
5.5 yrs (base)	69.95	78.19	—
7.0 yrs (high)	70.08	78.33	+0.13 yrs

The tight variance of ± 0.13 years confirms that the overall e metric is not materially sensitive to assumptions made about the open-ended tail. This demonstrates that the primary driver of total life expectancy in the Vietnamese population remains mid-life mortality, rather than the extreme tail behavior observed at ages 85+.

5. Conclusion

This analysis successfully establishes a robust, population-level mortality baseline for Vietnam across two critical reference years. By translating raw data into functional actuarial metrics, the project underscores how rapid socio-economic shifts actively reshape a nation’s risk profile. The pronounced young male accident hump, the distinct rectangularization of the survival curve, and the massive multi-year expansion in life expectancy all point to one reality: modern pricing models can no longer rely on static or gender-neutral assumptions without exposing the insurer to severe adverse selection or under-reserving risks.

Limitations

- Reliance on Population-Level Data: The analysis utilizes general population data rather than insured lives experience. An insured lives study would be required to accurately calibrate these tables to a specific company’s portfolio.
- Absence of Mortality Improvement Projections: The pricing implications discussed in Section 4.3 are strictly qualitative. Incorporating a proper mortality improvement scale (e.g., CMI or SOA MP-2021) is necessary for actual production pricing.
- Data Granularity Constraints: The use of five-year age groupings limits the precision required for single-age pricing models, necessitating future interpolation for finer product development.

References

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